

# **Chapter 3.A**

## **Compounds and the Mole**

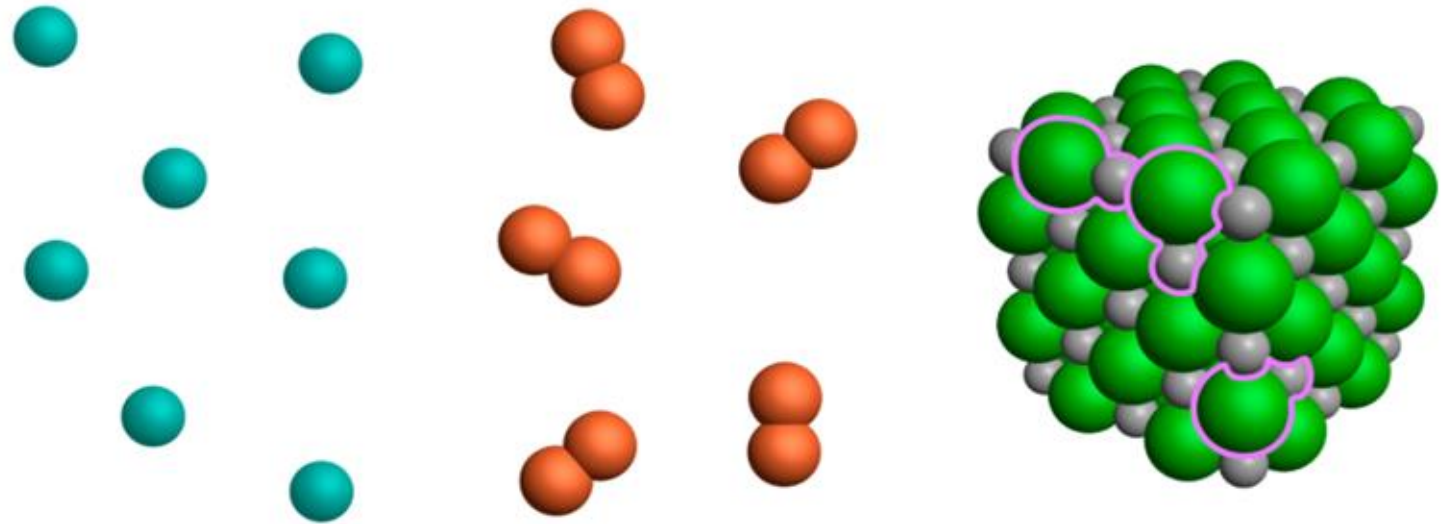
Chemical Principles I

CHEM 1211K

1. Interpret chemical formulas.
2. Write chemical formulas to represent simple inorganic chemical compounds.
3. Name simple inorganic chemical compounds.
4. Calculate the number of moles from the number of particles and vice versa.
5. Calculate the mass of 1 mole of a substance based on its chemical formula and atomic masses from the periodic table.
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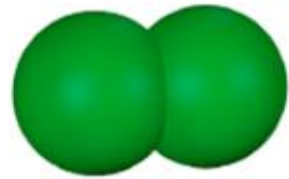
- **Chemical formulas** combine chemical symbols with subscripted numbers, parentheses, and other symbols to represent chemical compounds.
- The group of atoms represented by a chemical formula is called a **formula unit**. Depending on the structure of the substance, the formula unit may correspond to a **molecule** or a **repeating unit** in an infinite lattice.

- Substances vary in the nature of bonding between atoms or ions and the corresponding submicroscopic picture.
  - Molecular (covalent) compounds**
  - Ionic compounds**



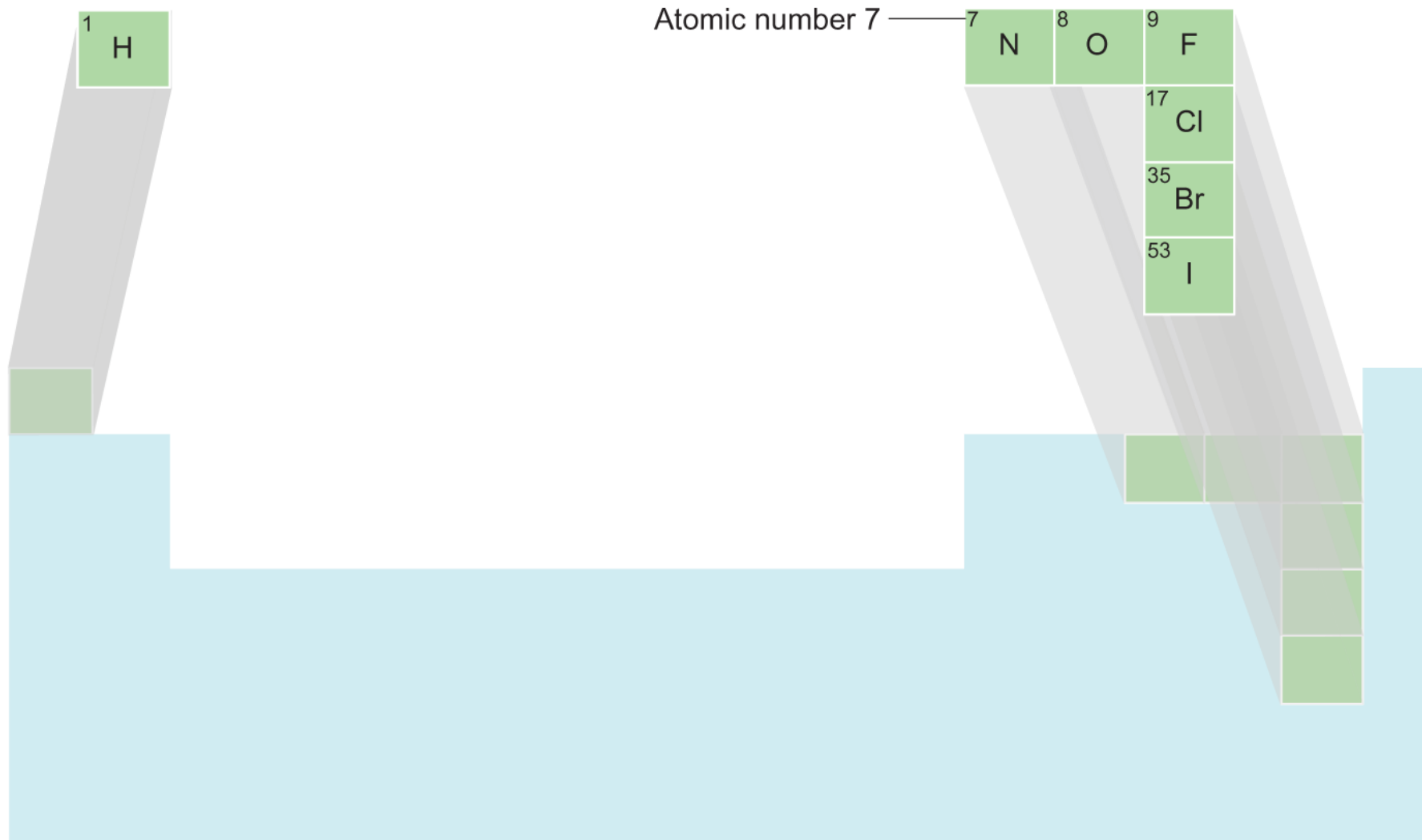
**Figure 3.1.** Atoms of argon ( $\text{Ar}$ ), molecules of dibromine ( $\text{Br}_2$ ), and an ionic lattice of sodium chloride ( $\text{NaCl}$ ).

- Some elements exist in nature in molecular form, as small molecules containing two or more of the same kind of atom bonded together.



**Figure 3.2.** Elemental chlorine ( $\text{Cl}_2$ ), phosphorus ( $\text{P}_4$ ), and sulfur ( $\text{S}_8$ ).

- Seven nonmetallic elements exist in nature as **diatomic** molecules.

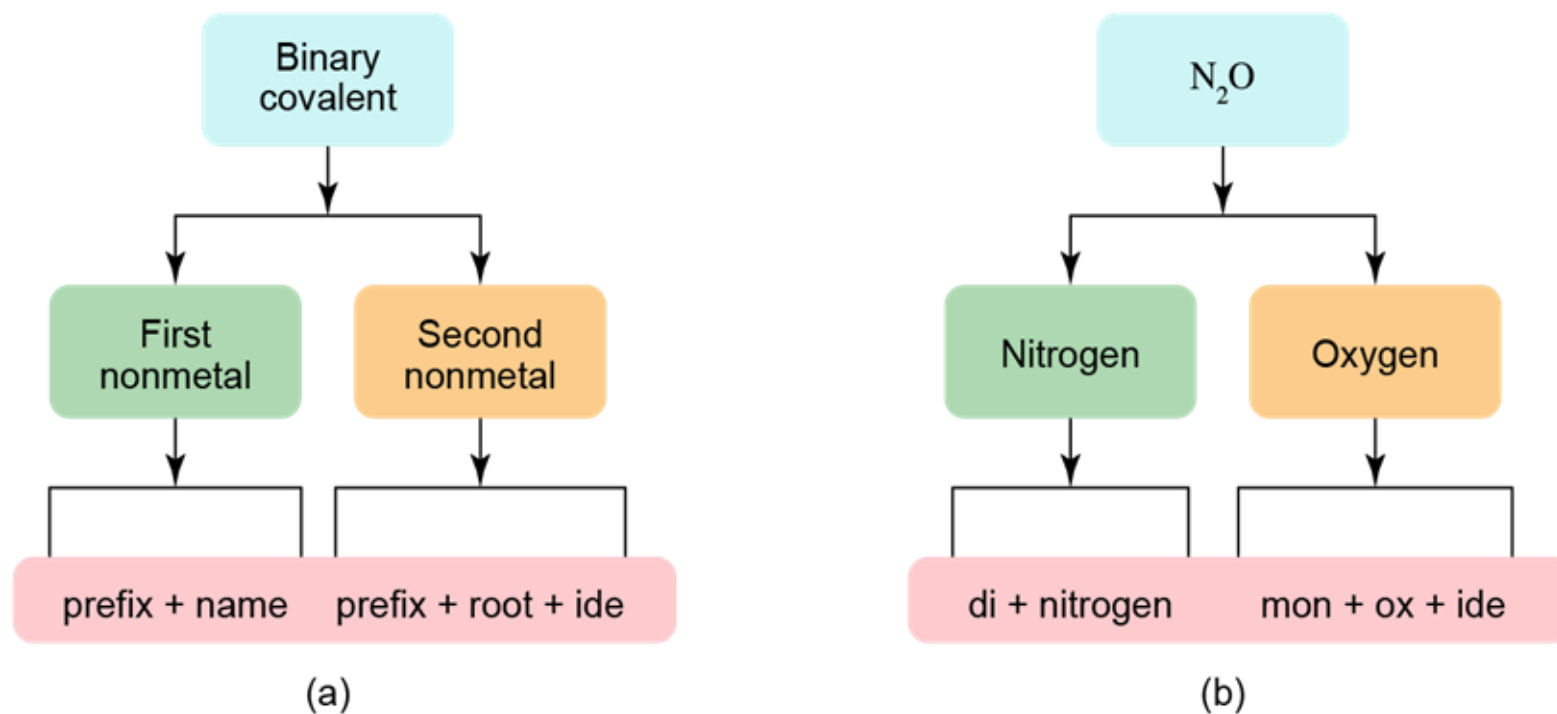


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**Figure 3.3.** The seven elements that exist in nature as diatomic molecules.

## 3.2 Naming Binary Covalent Compounds

- Molecular (covalent) compounds are composed of two or more *nonmetals*; **binary** covalent compounds contain only two elements.
- List the first nonmetal with a prefix indicating the number of atoms of that type in the molecule.
  - After a space, list the second nonmetal with a similar prefix, replacing the ending of the element name with *-ide*.



**Figure 3.4.** (a) A general scheme for naming binary covalent compounds and (b) an example.

| Prefix | Number of Atoms per Molecule |
|--------|------------------------------|
| mono   | 1                            |
| di     | 2                            |
| tri    | 3                            |
| tetra  | 4                            |
| penta  | 5                            |
| hexa   | 6                            |
| hepta  | 7                            |
| octa   | 8                            |
| nona   | 9                            |
| deca   | 10                           |

What is the chemical formula for disulfur hexoxide?

A. SO

B.  $S_6O_2$

C.  $SO_6$

D.  $S_2O_6$

E.  $S_2O_7$



### 3.3 Formulas for Ionic Compounds | 1 of 4

- **Ions** are species with a lack or excess of electrons, which therefore have net positive charge (**cations**) or negative charge (**anions**).
- You should become familiar with the charges of stable monatomic ions of the elements in the main group (1, 2, 13 – 18). *Note the pattern!*

|             |          |   |              |             |              |              |              |              |                  |              |              |          |  |              |         |          |          |          |          |          |          |          |  |
|-------------|----------|---|--------------|-------------|--------------|--------------|--------------|--------------|------------------|--------------|--------------|----------|--|--------------|---------|----------|----------|----------|----------|----------|----------|----------|--|
| 1           |          |   |              |             |              |              |              |              |                  |              |              |          |  |              |         |          | 18       |          |          |          |          |          |  |
| 1A          | 2        |   |              |             |              |              |              |              |                  |              |              |          |  |              |         |          |          | 8A       |          |          |          |          |  |
| H<br>+1, -1 | 2A       |   |              |             |              |              |              |              |                  |              |              |          |  |              |         |          |          | 13<br>3A | 14<br>4A | 15<br>5A | 16<br>6A | 17<br>7A |  |
| Li<br>+1    | Be<br>+2 |   |              |             |              |              |              |              |                  |              |              |          |  |              |         |          |          |          |          | N<br>-3  | O<br>-2  | F<br>-1  |  |
| Na<br>+1    | Mg<br>+2 | 3 | 4            | 5           | 6            | 7            | 8            | 9            | 10               | 11           | 12           | Al<br>+3 |  |              | P<br>-3 | S<br>-2  | Cl<br>-1 |          |          |          |          |          |  |
| K<br>+1     | Ca<br>+2 |   | Ti<br>+2, +3 | V<br>+2, +3 | Cr<br>+2, +3 | Mn<br>+2, +3 | Fe<br>+2, +3 | Co<br>+2, +3 | Ni<br>+2, +3, +4 | Cu<br>+1, +2 | Zn<br>+2     |          |  |              |         | Se<br>-2 | Br<br>-1 |          |          |          |          |          |  |
| Rb<br>+1    | Sr<br>+2 |   |              |             |              |              |              |              |                  | Pd<br>+2, +4 | Ag<br>+1     | Cd<br>+2 |  | Sn<br>+2, +4 |         |          | I<br>-1  |          |          |          |          |          |  |
| Cs<br>+1    | Ba<br>+2 |   |              |             |              |              |              |              |                  | Pt<br>+2, +4 | Au<br>+1, +3 | Hg<br>+2 |  | Pb<br>+2, +4 |         |          |          |          |          |          |          |          |  |
| Fr<br>+1    | Ra<br>+2 |   |              |             |              |              |              |              |                  |              |              |          |  |              |         |          |          |          |          |          |          |          |  |

Figure 3.7

- **Ionic compounds** are composed of **metal** and **nonmetal** atoms in ionic form; **binary** ionic compounds consist of one type of cation and one type of anion. Either or both may be polyatomic.
- Ionic compounds are electrically neutral overall; we can use this idea to deduce chemical formulas or charges.

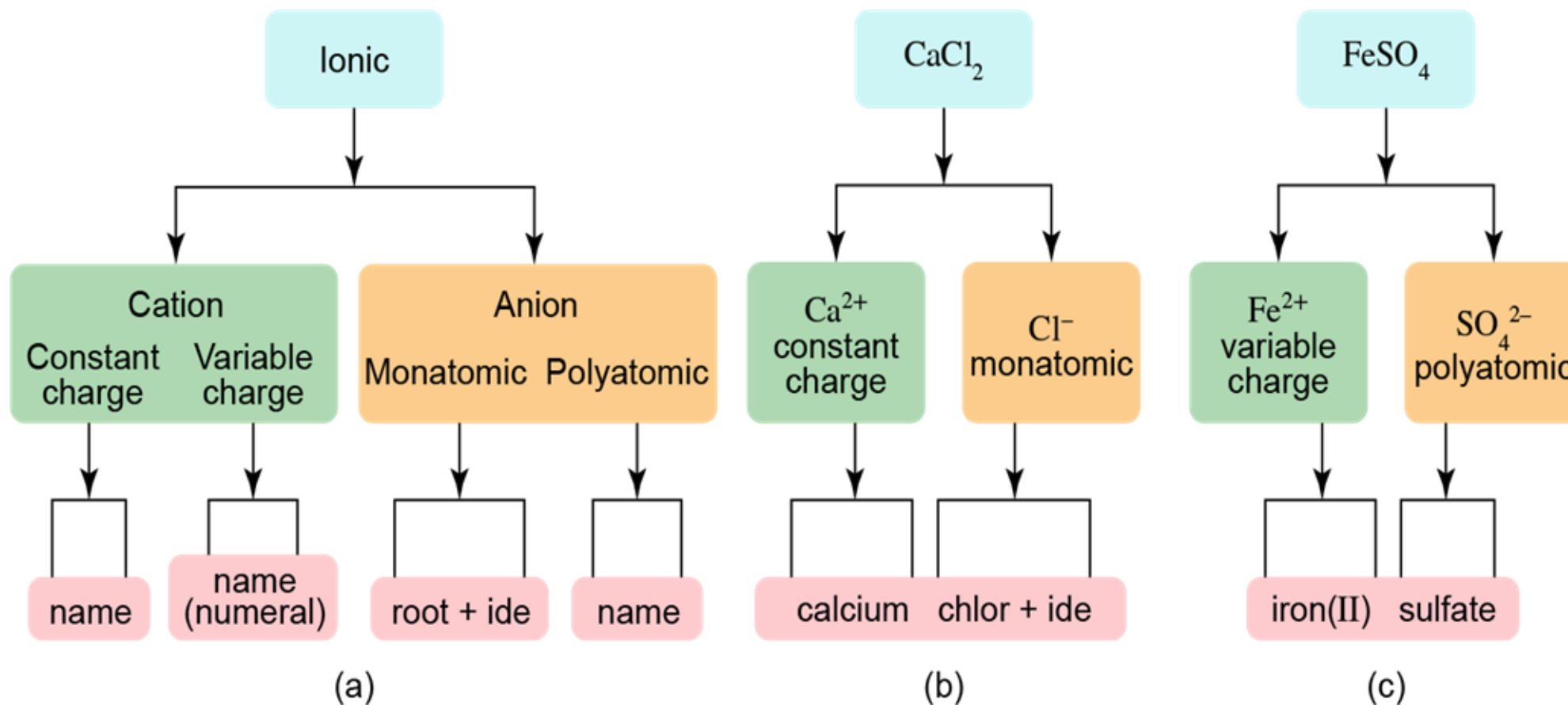
### 3.3 Formulas for Ionic Compounds | 3 of 4

- **Polyatomic ions** contain more than one type of atom; bonds *within* a polyatomic ion are covalent and the net charge may be positive (lack of electrons) or negative (excess of electrons).

| Name                  | Formula                                 | Name         | Formula                      |
|-----------------------|-----------------------------------------|--------------|------------------------------|
| acetate               | $\text{C}_2\text{H}_3\text{O}_2^-$      | hypochlorite | $\text{ClO}^-$               |
| ammonium              | $\text{NH}_4^+$                         | chlorite     | $\text{ClO}_2^-$             |
| carbonate             | $\text{CO}_3^{2-}$                      | chlorate     | $\text{ClO}_3^-$             |
| cyanide               | $\text{CN}^-$                           | perchlorate  | $\text{ClO}_4^-$             |
| hydrogen carbonate    | $\text{HCO}_3^-$                        | chromate     | $\text{CrO}_4^{2-}$          |
| hydrogen phosphate    | $\text{HPO}_4^{2-}$                     | dichromate   | $\text{Cr}_2\text{O}_7^{2-}$ |
| hydrogen sulfate      | $\text{HSO}_4^-$                        | permanganate | $\text{MnO}_4^-$             |
| hydroxide             | $\text{OH}^-$                           |              |                              |
| nitrite   nitrate     | $\text{NO}_2^-$   $\text{NO}_3^-$       |              |                              |
| peroxide              | $\text{O}_2^{2-}$                       |              |                              |
| phosphite   phosphate | $\text{PO}_3^{3-}$   $\text{PO}_4^{3-}$ |              |                              |
| sulfite   sulfate     | $\text{SO}_3^{2-}$   $\text{SO}_4^{2-}$ |              |                              |

### 3.4 Naming Binary Ionic Compounds | 4 of 4

- Because ionic compounds are electrically neutral and ion names include information about charge, prefixes are *omitted* in names of binary ionic compounds.
- Ions with variable charge require a Roman numeral in parentheses after the name of the ion.

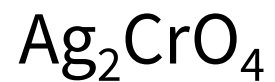


**Figure 3.10.** (a) A general scheme for naming binary ionic compounds and (b, c) two examples.

How many oxide ions are required to form an ionic compound with the stable monatomic ion formed by calcium?

How many sulfite ions are required to form a neutral ionic compound with the ion formed when vanadium has 20 electrons?

Name the chemical compounds below.



# **Chapter 3.B**

## Compounds and the Mole

Chemical Principles I

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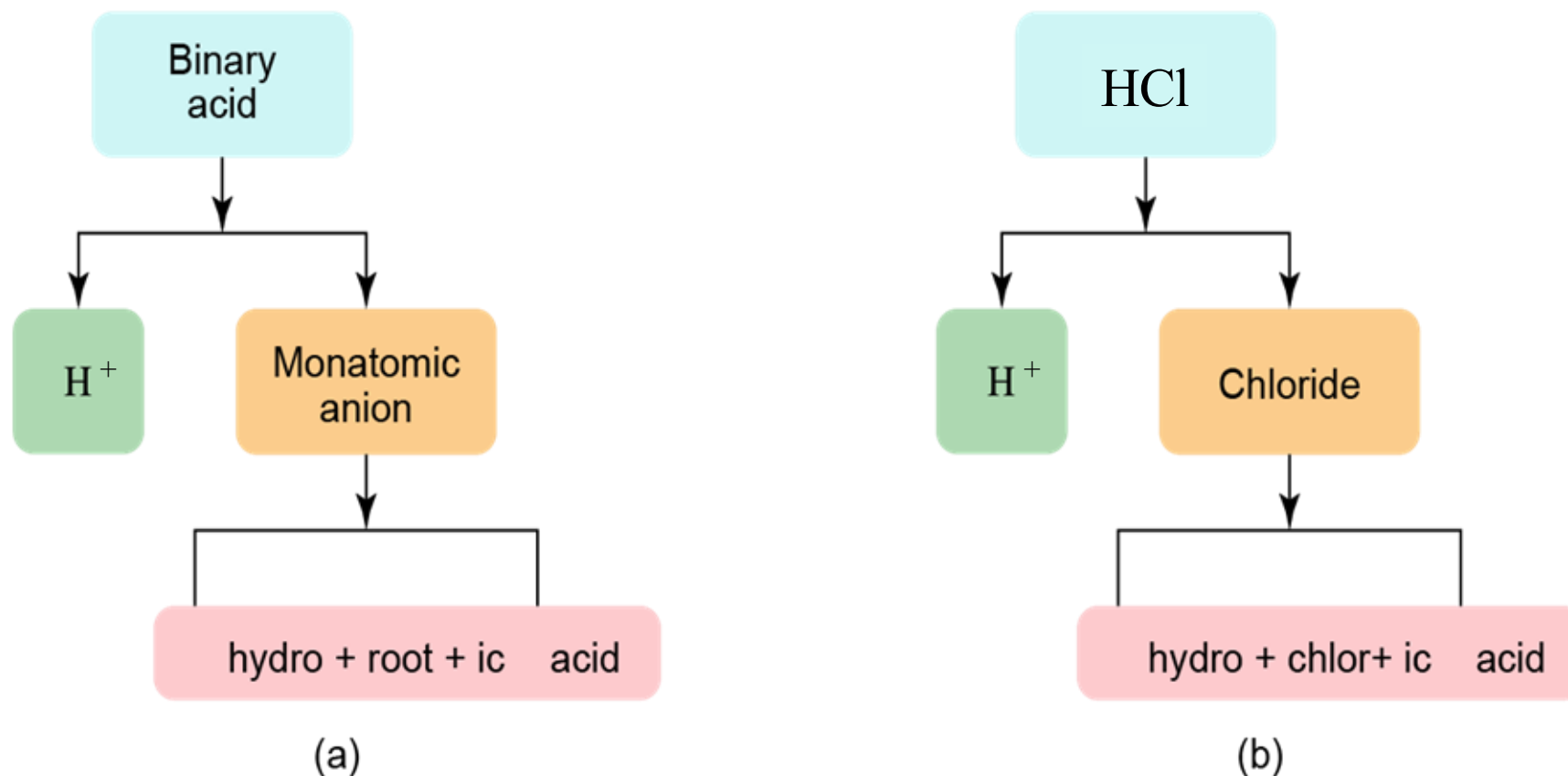


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- **Acids** are a special class of hydrogen-containing covalent compounds that, when dissolved in water...
- **Ionizable hydrogens** are those within an acid that can be lost as  $\text{H}^+$ .

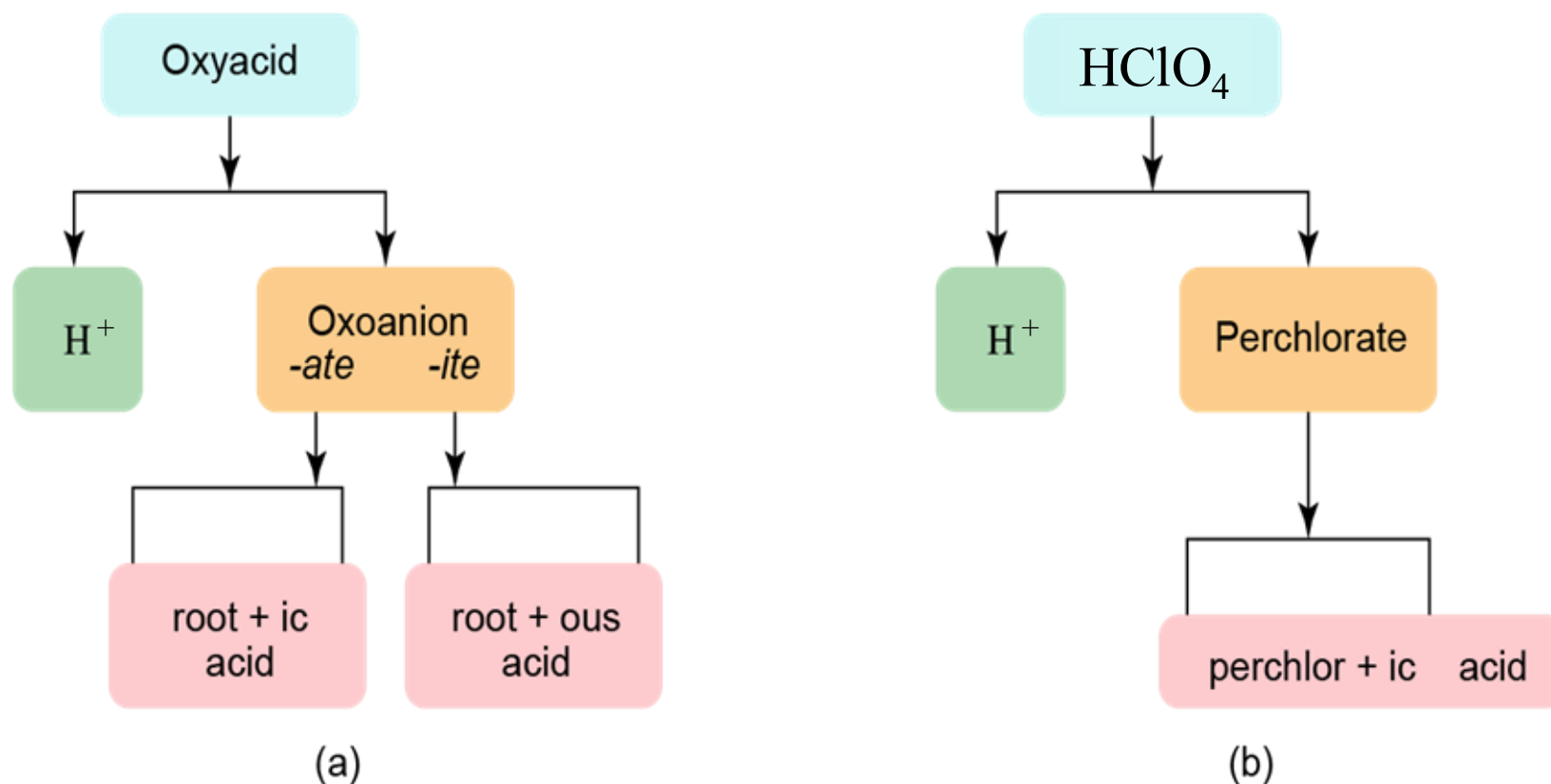
- We will be concerned with two types of acids: **binary acids  $H_nX$**  and **oxyacids  $H_nXO_m$** .
- **Examples**
  - HCl is a binary acid.
  - $HClO_4$  is an oxyacid.

- **Binary acids** contain hydrogen and one other element, typically from group 16 or 17.



**Figure 3.13.** (a) A general scheme for naming binary acids and (b) an example.

- Oxyacids** contain protons ( $\text{H}^+$ ) bonded to polyatomic anions containing oxygen (oxyanions  $\text{XO}_m^{n-}$ )



**Figure 3.14.** (a) A general scheme for naming oxyacids and (b) an example.

What is the name of the acid  $\text{HNO}_2$ ?

- A. hydronitric acid
- B. hydronitrous acid
- C. nitrous acid
- D. nitric acid

What is the name of the acid  $\text{HBrO}_4$ ?

## 3.6 Review of Nomenclature

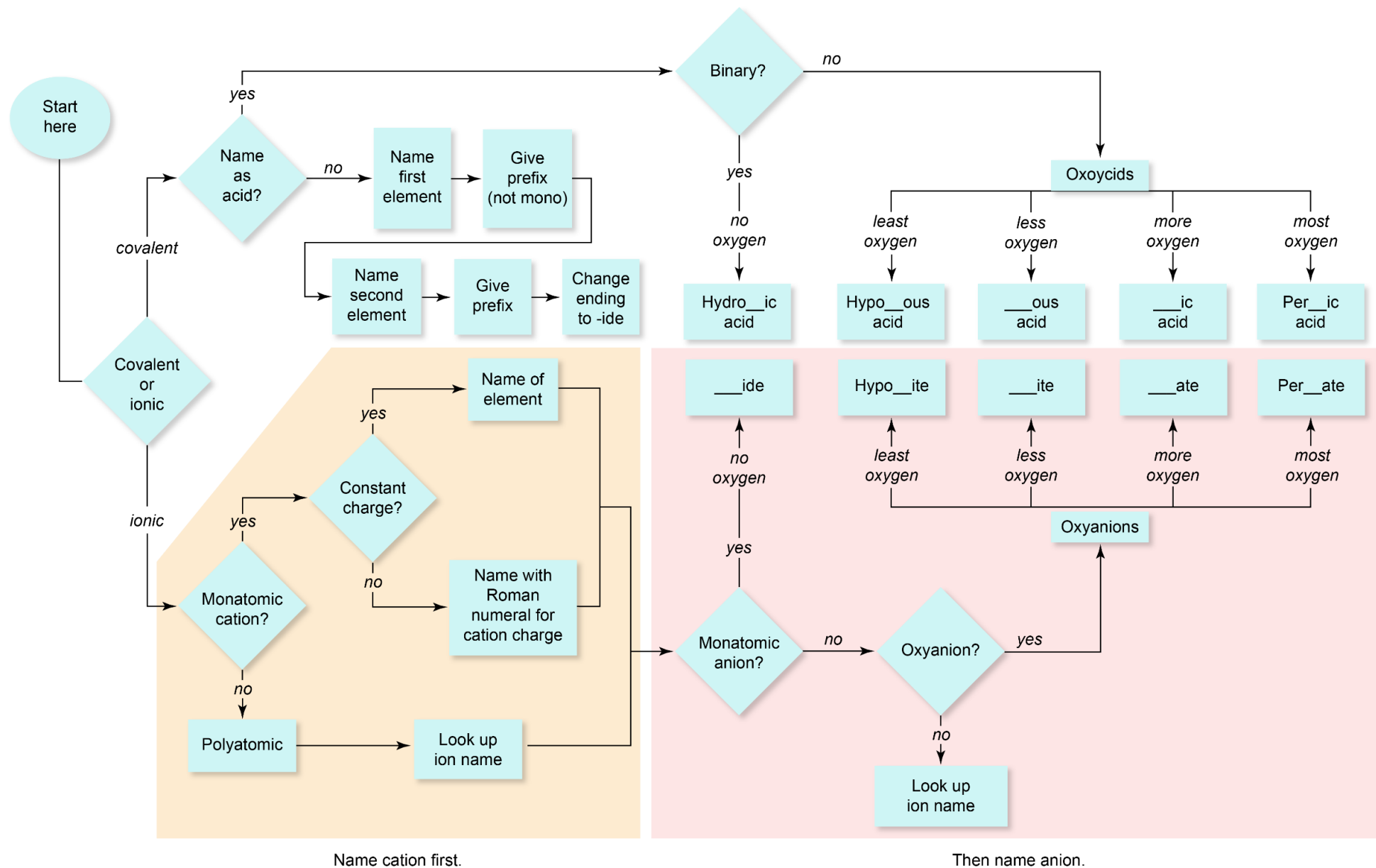


Figure 3.15



- The **mole (mol)** is a unit used to count the huge number of molecules or formula units in macroscopic samples.
  - Defined as the number of atoms in exactly 12.0 grams of carbon-12.
  - This definition ensures that 1 mol of a substance with formula mass  $M$  atomic mass units weighs  $M$  grams.
  - One mole contains **Avogadro's number ( $N_A$ )** of molecules, formula units, or other submicroscopic entity:  
 $6.022 \times 10^{23} \text{ mol}^{-1}$ . [This is really, really big!](#)

Number  
of moles

Avogadro's  
number

Number of  
formula units

- The **atomic mass** of each element is the average mass of one atom, expressed in atomic mass units (u or amu).
- The formula mass of a compound in atomic mass units (**molecular weight**) is equal to the mass of one mole of the compound (**molar mass**) in grams per mole.
- To find the molar mass of a compound, we simply add up the atomic masses of all atoms in one formula unit of the compound.
- **Example.** Determine the molar mass of  $\text{C}_2\text{H}_6\text{O}$ .  
Relevant molar masses: C = 12.01 g/mol; H = 1.01 g/mol; O = 16.00 g/mol.

- Molar mass (g/mol) enables us to...
  - Determine the number of moles corresponding to a given mass of substance
  - Determine the mass of a given number of moles of substance

What is the count of oxygen atoms in a sample of 2.67 moles of hydrogen peroxide ( $\text{H}_2\text{O}_2$ )?

Enter your response as the base-10 logarithm of this number to the nearest 0.01.

How many atoms of oxygen are present in 255 g of potassium dichromate? [Determine the chemical formula first!](#)

Enter your response as the base-10 logarithm of this number to the nearest 0.01.

Calculate the molar mass in g/mol of a covalent compound if the mass of a single molecule is  $5.34 \times 10^{-23}$  g.

Enter your response to the nearest 0.01 g/mol.

# **Chapter 3.C**

## **Compounds and the Mole**

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- Some notes on vocabulary...
  - **Formula mass (weight)** is a generic term that may refer to an atoms, molecule, or formula unit.
  - **(Average) atomic mass (weight)** refers specifically to atoms.
  - **Molecular mass (weight)** refers specifically to molecules, although this term is sometimes loosely—and incorrectly—applied to formula units of ionic compounds.
  - **Molar mass** refers to the mass of one mole of atoms, molecules, or formula units.
- We'll use the last two terms most commonly, although the distinction between molecular and ionic compounds will remain important throughout CHEM 1211K and 1212K!

## 3.9 Percent Composition



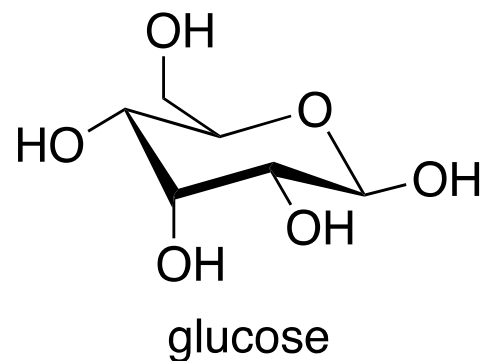
- **Percent composition** refers to the *mass* percentage of each element within a compound.
- If the chemical formula is known, percent composition can be deduced from the formula and the average atomic masses of the constituent elements.
- **Example.** What are the mass percentages of carbon and hydrogen in methane ( $\text{CH}_4$ )? The molar mass of carbon is 12.011 g/mol and that of hydrogen is 1.008 g/mol.

- Mass percent composition only enables us to determine the formula of a compound as the lowest whole-number ratio of the elements.
  - **Chemical formula** is a general term referring to the numbers and types of the atoms in a compound.
  - **Empirical formula** includes the lowest or simplest whole-number ratios of the elements in the compound.
  - **Molecular formula** refers to the composition of a single molecule or formula unit of the substance.

- The empirical and molecular formulas of a compound frequently differ.  
Consider ethylene, which has the molecular formula  $\text{C}_2\text{H}_4$ .
- To determine an empirical formula from given mass percentages of the elements...
  1. Convert percentage data into mass data, assuming a basis of 100.0 g of substance.
  2. Determine the number of moles corresponding to the mass of each element.
  3. Divide the moles of all elements by the smallest number of moles.
  4. If the resulting numbers are not all whole numbers, scale up by an appropriate factor to make all whole numbers.
  5. Insert the resulting numbers as subscripts into the chemical formula.

## 3.11 Molecular Formulas

- The **molecular formula** contains the composition of one molecule of a molecular compound. Molecular formula is necessarily a multiple of empirical formula (the multiplier may be 1).
- For example, one molecule of glucose contains 6 C atoms, 12 H atoms, and 6 O atoms. The molecular formula is  $C_6H_{12}O_6$ . What is the empirical formula?



### 3.11 Determining a Molecular Formula



**Example.** Vitamin C contains the elements C, H, and O. It is known to contain 40.9% C, 54.5% O, and 4.58% H by mass. The mass of 0.500 moles of vitamin C is 88.07 g. How many atoms of carbon are in a molecule of vitamin C?

- First, we'll use the given mass percentages to determine the empirical formula of vitamin C.
- Then, we'll deduce the molar mass, formula mass, and *molecular* formula.

Determine the empirical formula of acetic acid, which consists of 40.00% carbon, 6.71% hydrogen, and 53.29% oxygen by mass.

A mass of 6.01 grams of acetic acid corresponds to 0.100 mol of this substance.  
What is the molecular formula of acetic acid?



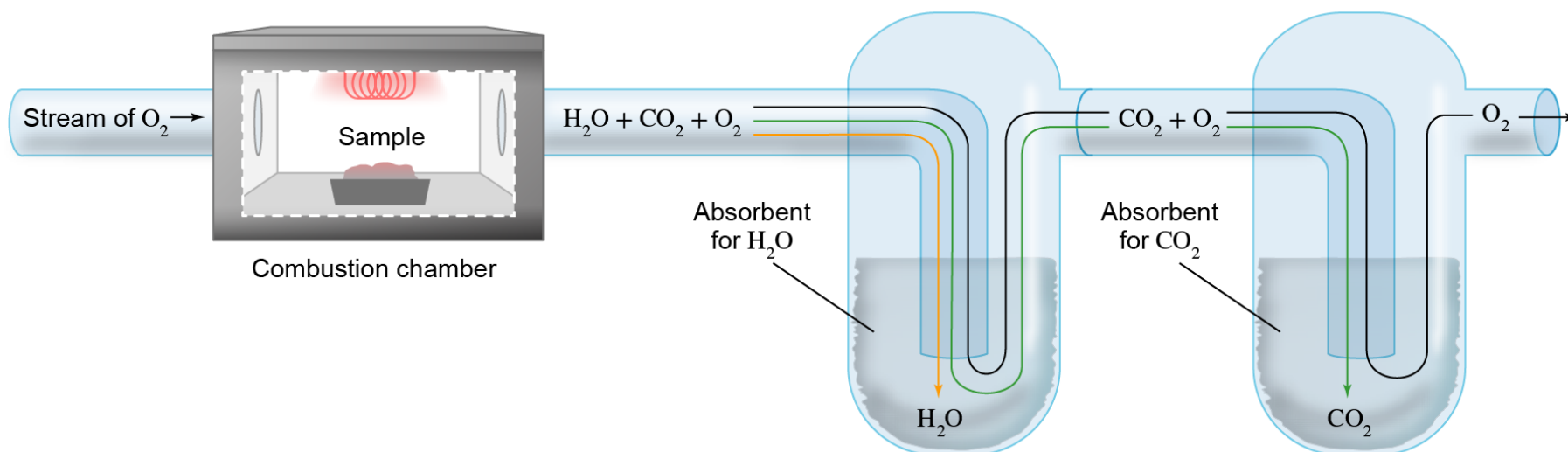
- We've already seen that subscripts in chemical formulas help us count atoms in one formula unit of a compound. They can also be used to count *moles* of an element in *one mole* of the compound.
- For example, consider theobromine ( $\text{C}_7\text{H}_8\text{N}_4\text{O}_2$ ), an active ingredient in chocolate. If you eat 1 mol of theobromine, you have consumed 7 mol of carbon atoms.

What is the mass of chromium (chromium *atoms*) in 225 g of potassium dichromate?

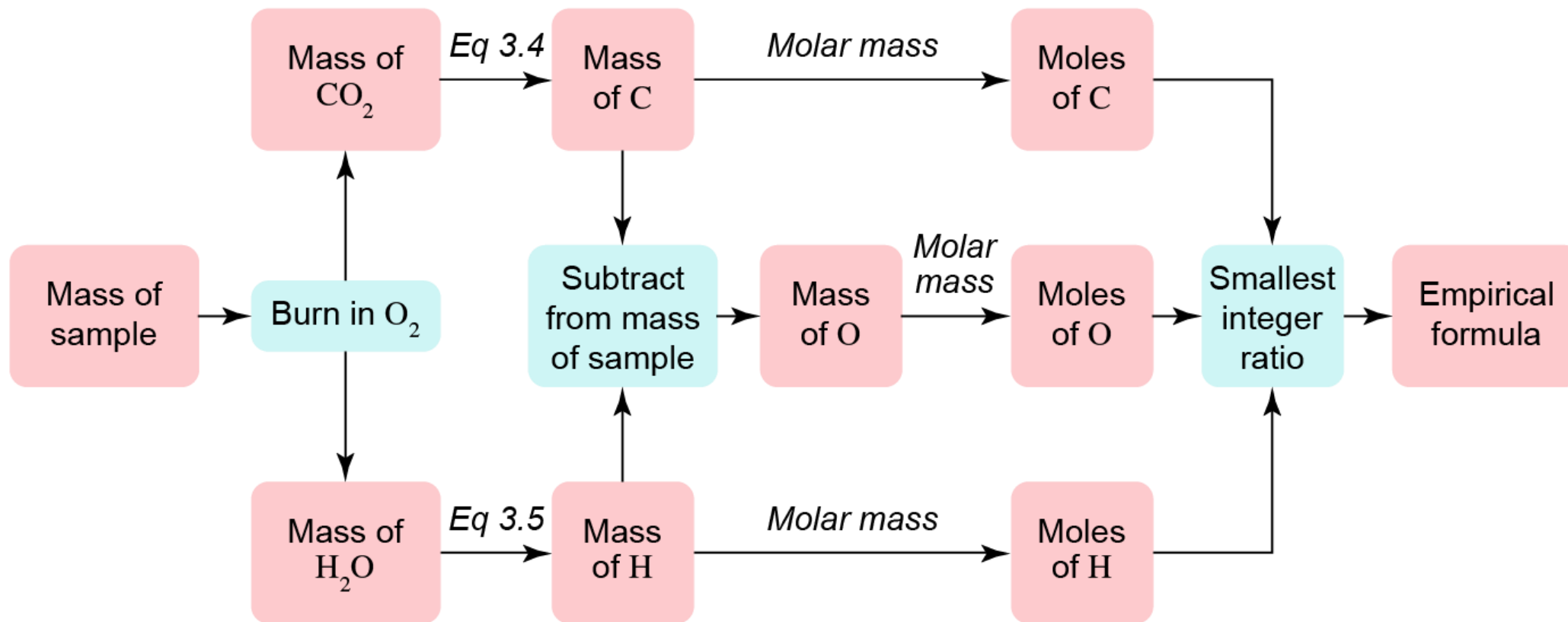
Determine the chemical formula first!

## 3.12 Combustion Analysis

- **Combustion analysis** is a laboratory technique used to determine empirical formula of compounds containing carbon, hydrogen, and perhaps a third element.
- A known mass of sample reacts with molecular oxygen ( $O_2$ ).



**Figure 3.23.** Apparatus for combustion analysis.



**Figure 3.24.** Flow chart for solving combustion analysis problems.

A 0.250 g sample of hydrocarbon undergoes complete combustion to produce 0.845 g of  $\text{CO}_2$  and 0.173 g of  $\text{H}_2\text{O}$ .  
What is the empirical formula of this compound?

- Some of the amazing things we can now do with compounds and the mole...
  1. Distinguish between molecular and ionic compounds using chemical formulas.
  2. Visualize molecular and ionic compounds on the submicroscopic level.
  3. Name binary covalent compounds; translate the name of a binary covalent compound into a chemical formula.
  4. Name binary ionic compounds; translate the name of a binary ionic compound into a chemical formula.
  5. Name binary acids; translate the name of a binary acid into a chemical formula.
  6. Name oxyacids; translate the name of an oxyacid into a chemical formula.
  7. Determine the molar mass of an element or compound from its chemical formula.
  8. Apply molar mass to “convert” between mass and moles.
  9. Infer the empirical formula of a compound from the mass percentages of the elements within that compound.
  10. Apply chemical formulas to infer numbers or masses of atoms within a compound.
  11. Use the results of a combustion analysis experiment to infer the empirical formula of a compound containing carbon, hydrogen, and perhaps one other element.